

Charge Transport Properties of Dielectrics Revealed by Isothermal Surface Potential Decay

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ABSTRACT

Revealing the charge transport properties of space grade high insulation materials can benefit the mitigation of electrostatic discharges (ESD) on spacecraft. The charge transport properties of polyimide are investigated by an isothermal surface potential decay (ISPD) experiment under a simulated space environment chamber. After irradiated the sample by an electron gun, the 2-D surface potential distributions are measured by a non-contact potential probe. From the surface potential decay curves, we obtain current density at steady state ISPD. Analyzing the steady state current density against surface potential, we find two regimes. One is Ohmic regime and another is Space Charge Limited Current (SCLC) regime, which are separated at around -950 V with the sample thickness of 27 μm at 298 K. Ohmic resistivity and effective charge carrier mobility are calculated from these two regimes, respectively. In addition, the trap density of polyimide is derived from the SCLC theory, when the sample is charged to high initial surface potential.

Index Terms—Charge transport properties, isothermal surface potential decay, Ohmic regime, polyimide, space charge limited current regime.

NOMENCLATURE

Symbol	Quantity	Unit
EF	electric field	V/m
E_{ec}	conduction band energy	eV
E_{eT}	trap energy of electrons	eV
j_{Ohmic}	Ohmic current density	A/m ²
j_{out}	current density flowed out from the insulator to the grounded electrode	A/m ²
j_{SCLC}	space charge limited current density	A/m ²
j_S	surface current density	A/m ²
L	thickness of the material	m
$n_{trapped}$	trapped electrons	m ⁻³
N_T	trap density	m ⁻³
$p_{trapping}$	probability of a charge captured by empty trap centers per time	s ⁻¹
q_e	electron charge	C
q_S	surface charge density on the irradiated surface	C/m ²
q_{free}	free charge density	C/m ³
$q_{trapped}$	trapped charge density	C/m ³
q_{total}	total charge density	C/m ³
q_{total_T}	Total electron density in the bulk in thermal and electrical equilibrium without applied voltage	C/m ³
R_S	surface resistivity of the insulator	Ω
t	elapsed time of ISPD experiment	s
t_{st}	the steady state of ISPD	s

T	temperature of the material	K
V	applied voltage on the material	V
V_T	transition voltage between Ohmic regime and SCLC regime	V
x	depth in material from the irradiated surface	m
ϵ_0	dielectric permittivity of vacuum	F/m
ϵ_r	relative permittivity of dielectric	
ϕ_s	surface potential of dielectric at $x=0$	V
μ	charge carrier mobility	m ² /V·s
μ_{eff}	effective charge carrier mobility	m ² /V·s
θ	the ratio of free charge density and total charge density	
ρ_{Ohmic}	Ohmic resistivity of material	$\Omega\cdot\text{m}$

1 INTRODUCTION

WITH the increasing requirement of large power (> 10 kW) and high operational voltage (>100 V) for satellites, the space society takes more attentions to the anomalies of power systems resulted by electrostatic discharges (ESD) [1-4]. 52 % of recent satellite failures are attributed to the malfunctions of power system; for example, a Geosynchronous Orbit (GEO) satellite, Tempo-2, suddenly lost 15 % of its total power due to a sustained arc in 1997 [5].

Operating in space, a satellite will be charged by plasma, energetic electrons and protons, X-rays, Ultraviolet rays,

and so on. Once the potential difference between insulators and conductors or vacuum exceeds a critical value, ESD may occur on the material surface, inside the insulator, or at the triple junction that consists of insulator, conductor, and plasma (or vacuum). The ESD at the triple junction of solar array is extremely harmful for the satellite. According to [6], 33 % of recent satellite anomalies are caused by the malfunction of solar arrays.

The potential difference built-up on the satellite that determines the arc rate depends not only on space environment, but also on the satellite structure and the dielectric properties of insulators. A significant property of insulators related to spacecraft charging and discharging is charge transport. Polyimide is widely used on spacecraft for its excellent electrical and thermal properties [7]. However, its conduction characteristic has not been fully understood. In present paper, we concentrate on the charge transport properties of polyimide.

The Ohmic resistivity of polyimide is so high that the American Society for Testing and Materials (ASTM) method cannot be used. A charge storage decay resistivity test method was developed by A. R. Frederickson *et al.* to study the Ohmic resistivity of high insulation materials [8-10]. Fitting the potential decay curve by the charge depolarization and conduction model, they found that the Ohmic resistivity of polyimide is several order higher than the results obtained by the ASTM method.

When the charging potential of insulator is larger than the transition voltage of the Ohmic regime, the charge transport might change into high field mechanisms [11, 12]. Space-Charge-Limited-Current (SCLC) seems to be one possible conduction mechanism for polyimide [13-16]. The SCLC regime may be more important in spacecraft charging and discharging than the Ohmic regime, because it determines charge leakage speed at high electric field, and pre-breakdown characteristics or electrical life [17, 18]. For the limitation of the ASTM method, Y. Murakami *et al.* studied the charge carrier mobility of polyimide by charge chromatography method [19], and G. M. Sessler *et al.* studied the electrical conduction phenomena of polyimide at high temperatures [13].

An important interrelated parameter of the SCLC regime is the trap density of insulators. It not only determines the current and the trapped charge depth distribution, but also affects the dielectric aging characteristic [20]. Several researchers have done some works on the trap density of polyimide by different methods, such as Thermally Simulated Current method, voltammetry, and so on [13, 21-25].

The purpose of the present paper is to acquire consistent transport parameters of polyimide. To do so, we developed an Isothermal Surface Potential Decay (ISPD) model to study the trap density, which is described in details in Section 2. The charge transport properties of polyimide are studied by ISPD experiment. We describe the experimental set-up in Section 3. In Section 4, we compare the experimental results with theory. From the current density-voltage characteristic at steady state ISPD, we find that the

charge transport of polyimide is divided into the Ohmic regime and the SCLC regime at around -950 V with the thickness of 27 μm at 298 K. The Ohmic resistivity and the charge carrier mobility are calculated from the current density-voltage characteristic. The trap density is also obtained from the ISPD model. In Section 5, we conclude the paper with suggestion of works to be done in near future.

2 THEORY OF ISOTHERMAL SURFACE POTENTIAL DECAY

When a dielectric is irradiated by low energy electrons for a short time, charges will be accumulated on the surface of the material and a surface potential will be built up. After the irradiation, the charges on the surface will be accelerated to flow into the material [25]. Some of the injected charges will be trapped into trap centers, some of them will drift through the conduction band to the other electrode. Part of thermally activated trapped charges will be detrapped to the conduction band and start hopping to the opposite electrode [26-28]. These processes have been proved by experimental results by utilizing pulsed electroacoustic (PEA) [29-31], Laser Induced Pressure Pulse (LIPP) [32, 33], and Pressure Wave Propagation (PWP) [34] techniques.

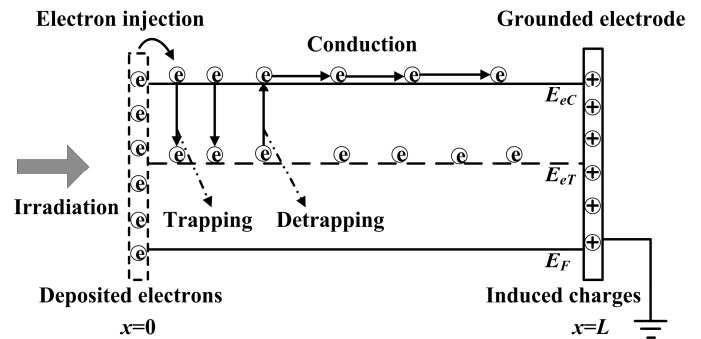


Figure 1. Schematic of charge injection into dielectric under high electrical stress, and charge trapping and detrapping process in the energy band. E_c is the conduction band; E_t is the trap energy; E_F is the Fermi level.

Figure 1 shows the schematic of the whole processes in the energy band. In the figure, the deposited electrons layer corresponds to the insulator surface and is defined as $x=0$. The anode corresponds to the grounded electrode and is defined as $x=L$. Electrons will be injected into and migrate in the upper half energy band [31]. In the mean time, the surface potential will decay with time, as shown in Figure 2 [8, 10].

The ISPD curve can be divided into three parts, which are resulted by different physical processes, respectively. Region I is dominated by charge trapping and detrapping processes. Region III is dominated by charge conduction. Region II is determined by the two processes [35]. Therefore, the ISPD property of insulating materials can be used to reveal the trapping, detrapping, and conduction characteristics.

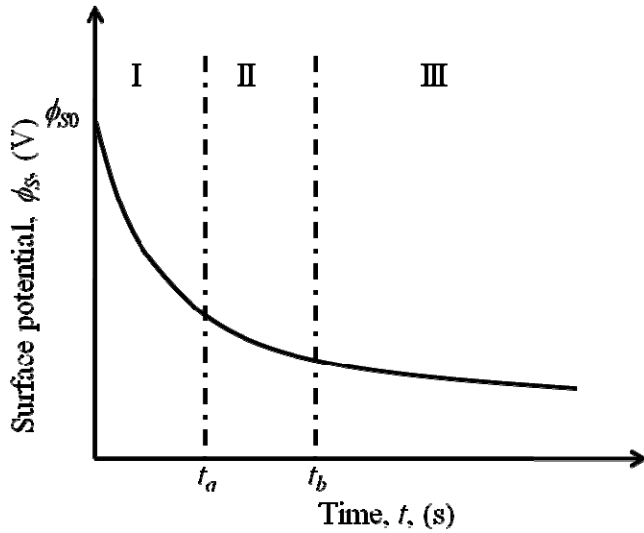


Figure 2. Schematic of Isothermal Surface Potential Decay (ISPD).

2.1 EVOLUTION OF SPACE CHARGE AND ELECTRIC FIELD PROFILES AT HIGH INITIAL SURFACE POTENTIAL

Figure 3a and 3b demonstrate that total electron density depth distribution and electric field depth distribution vary with decay time at high initial surface potential. The total electron density profiles reflect the charge transport in the insulator. The progressive migration of free electrons in the bulk generates an electron fraction in trap centers. The electrons inside the material then distort the electric field in the vicinity of the irradiated surface. The decrease in time of the electric field at the deposited electron layer causes a reduction of the subsequent injected electron quantity, which leads to the descend of surface potential decay speed.

The space charge profiles in the material can be utilized to calculate the trap density. In the following, we derive the steady state trapped charge depth distribution in the bulk under high initial surface potential. The current density with a constant charge carrier mobility neglecting charge diffusion can be interpreted by the conduction equation.

$$j_c = q_{free}(x) \cdot \mu \cdot EF(x) \quad (1)$$

where, where, j_c is the current density in A/m²; q_{free} is the free electron density in C/m³; μ is the charge carrier mobility in m²/V·s; EF is the electric field in V/m.

The electric field in the dielectric is determined by Poisson equation,

$$\frac{dEF(x)}{dx} = \frac{q_{total}(x) - q_{total_T}(x)}{\epsilon_0 \cdot \epsilon_r} \quad (2)$$

where, q_{total} is the total electron density in the material under electrical stress in C/m³; q_{total_T} is the total electron density in the bulk in thermal and electrical equilibrium without applied voltage in C/m³.

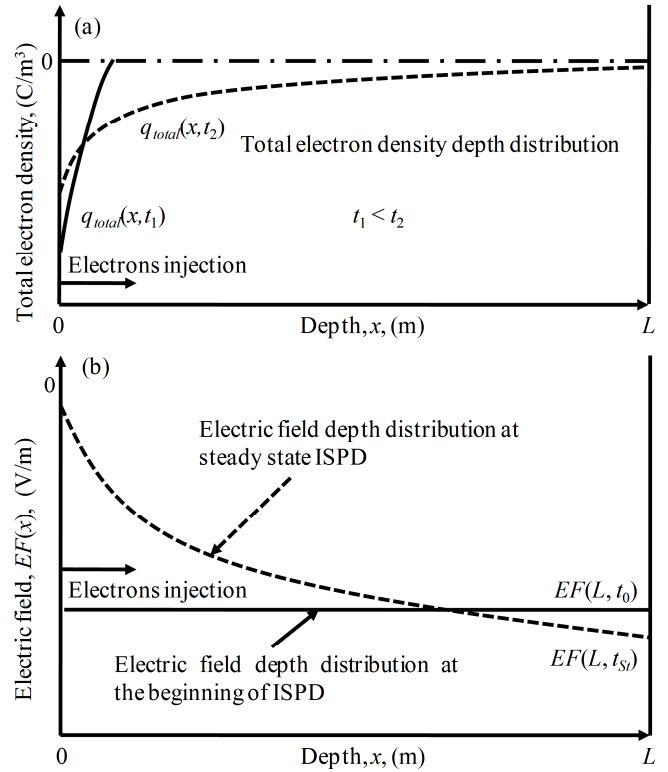


Figure 3. (a) Schematic of total electron density depth distribution at the beginning and the steady state of ISPD; (b) Schematic of potential depth distribution evolution of ISPD process.

For high insulating material, q_{total_T} is usually very small. The inequality, $q_{total_T} \ll q_{total}$, is assumed for polyimide under high initial surface potential. We set θ as the ratio of the free electron density and the total electron density, $\theta = q_{free}(x)/q_{total}(x)$, which is independent on the position x for shallow traps. The boundary condition of equations (1) and (2) is

$$EF(x)|_{x=L} = EF(L) \quad (3)$$

where, $EF(L)$ is the electric field at the grounded electrode in V/m. M. A. Lampert established a inequality for the electric field at the grounded electrode [26].

$$\phi_s/L < EF(L) < 2 \cdot \phi_s/L \quad (4)$$

where, ϕ_s is the surface potential of the irradiated insulator in V.

When substituted equation (2) into (1), the solution is

$$EF(x) = - \left(\frac{2 \cdot j_c \cdot (x-L)}{\epsilon_0 \cdot \epsilon_r \cdot \mu \cdot \theta} + EF(L)^2 \right)^{1/2} \quad (5)$$

Then we can get the trapped charge depth distribution, $q_{trapped}(x) = (1-\theta) \cdot q_{total}(x)$, by substituting equation (5) into (2).

$$q_{trapped}(x) = \frac{(\theta-1) \cdot j_c}{\theta \cdot \mu} \cdot \left(\frac{2 \cdot j_c \cdot (x-L)}{\epsilon_0 \cdot \epsilon_r \cdot \theta \cdot \mu} + EF(L)^2 \right)^{-1/2} \quad (6)$$

2.2 TRAP DENSITY

Considering the first order kinetics of trapped charges at time t , we can get the following equation.

$$\frac{dn_{trapped}(x,t)}{dt} = (N_T - n_{trapped}(x,t)) \cdot p_{trapping} \quad (7)$$

where, $n_{trapped}(t)$ is the trapped charges at time t in m^{-3} , $dn_{trapped}(t)/dt$ is the captured charges at time t with an increase of dt in $m^{-3} \cdot s^{-1}$, and N_T is the trap density in m^{-3} . $p_{trapping}$ is the probability of a charge captured by empty trap centers per time in s^{-1} .

We rearrange and integrate the equation (7).

$$N_T - n_{trapped}(x,t) = C \cdot \exp\left[-\int p_{trapping} \cdot dt\right] \quad (8)$$

Considering the initial condition at $t=0$,

$$n_{trapped}(x,t)\big|_{t=0} = 0 \quad (9)$$

we can solve equation (8).

$$n_{trapped}(x,t) = N_T - N_T \cdot \exp(-p_{trapping} \cdot t) \quad (10)$$

When the ISPD reaches steady state, the elapsed trapping times at $x = 0$ is long enough that we can neglect the second part at the right side of equation (10). Consequently, we obtain the trapped trapped electrons at $x = 0$.

$$q_{trapped}(0, t_{St}) = -q_e \cdot N_T \quad (11)$$

From equation (5), the trapped electrons at can also be obtained.

$$q_{trapped}(0, t_{St}) = \frac{(\theta-1) \cdot j_c}{\theta \cdot \mu} \cdot \left(\frac{-2 \cdot j_c \cdot L}{\varepsilon_0 \cdot \varepsilon_r \cdot \theta \cdot \mu} + EF(L)^2 \right)^{-1/2} \quad (12)$$

Assuming that θ is small enough so that we can approximate $(1-\theta)$ by unity in equation (12). Consequently, we obtain the trap density by equations (11) and (12) at the steady state ISPD.

$$N_T = \frac{j_c}{q_e \cdot \theta \cdot \mu} \cdot \left(\frac{-2 \cdot j_c \cdot L}{\varepsilon_0 \cdot \varepsilon_r \cdot \theta \cdot \mu} + EF(L)^2 \right)^{-1/2} \quad (13)$$

3 EXPERIMENTAL SETUP

A 27 μm thick polyimide film was used as samples. Copper was used as substrates. The polyimide film was fixed by copper tape that has enough conductance to transport surface charges, as shown in Figure 4. The uncovered area of the sample was a square of 4 cm length.

Before we carried out ISPD experiments, the material was baked in an oven at 373 K for 10 hours to eliminate the humidity and charges in the bulk. The sample was, then, put into a vacuum chamber with 0.6 m in diameter and 0.9 m in length, as shown in Figure 5. The vacuum chamber was evacuated by a Turbo-molecular pump (Osaka vacuum, 500 1/s) baked by a mechanical pump, and its pressure was kept below 5×10^{-4} Pa during the ISPD experiment. After keeping the sample in vacuum for 24 hours, we irradiated the sample

by an electron beam gun mounted on the top of the chamber. A double-layer grounded aluminum plate with a gap of 20 mm in between and two coaxial holes of 10 mm in diameter was used to allow electron beam to hit on the center of the sample. The irradiated electron beam energy was 3 keV with an emission current of 30 μA . The electron beam was shut down after 3 minutes of irradiation.

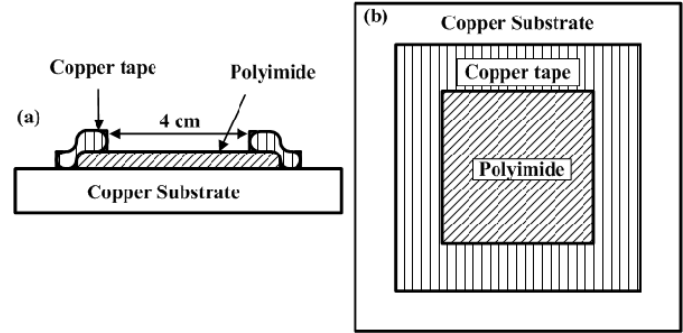


Figure 4. Schematic of experimental samples, (a) front view; (b) top view.

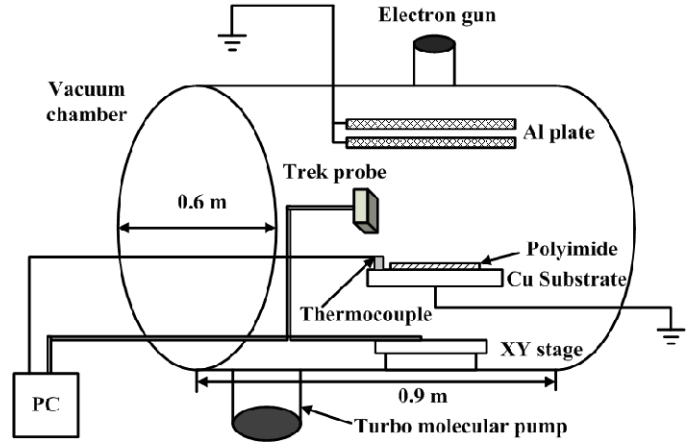


Figure 5. Experimental setup of isothermal surface potential decay.

A non-contact potential probe (TREK, model 341B) was mounted on a motor driven X-Y stage to measure the 2D distribution of surface potential of polyimide from 4 mm top. The Probe could scan an area of 50 mm $\times$ 50 mm within 120 seconds with 1 mm resolution. The probe returned to its origin, 10 cm away from the sample at the end of the scanning to avoid the influence of the probe on the sample. Two thermocouples were used to measure the temperature of the sample. The surface potential and temperature data of the dielectric were acquired by a data acquisition board, and recorded by two LABVIEW® programs.

4 RESULTS AND DISCUSSION

The temperatures of the sample during the ISPD experiment are demonstrated in Figure 6. Figure 6 indicates that the temperatures of the sample were stable in the vacuum chamber during the experiment, and were around 298 K.

The 2D surface potential distributions of electron irradiated polyimide were measured from the beginning ($t = 0$ h) to the

end of the ISPD experiment. Only two data are shown in Figure 7 (at $t = 0$ h) and Figure 8 (at $t = 94.8$ h). The surface potential at the irradiation center at $t = 0$ h was -2056 V and at the end ($t = 94.8$ h) was -1441 V.

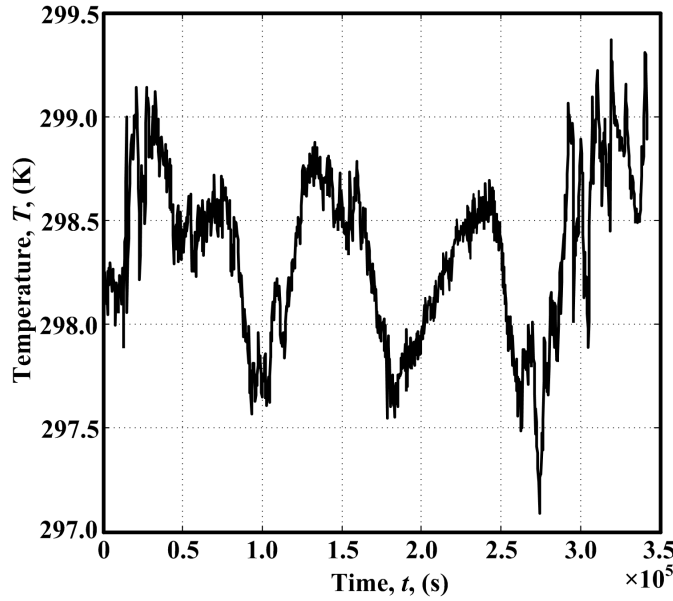


Figure 6. Temperatures of the sample during the surface potential decay experiment.

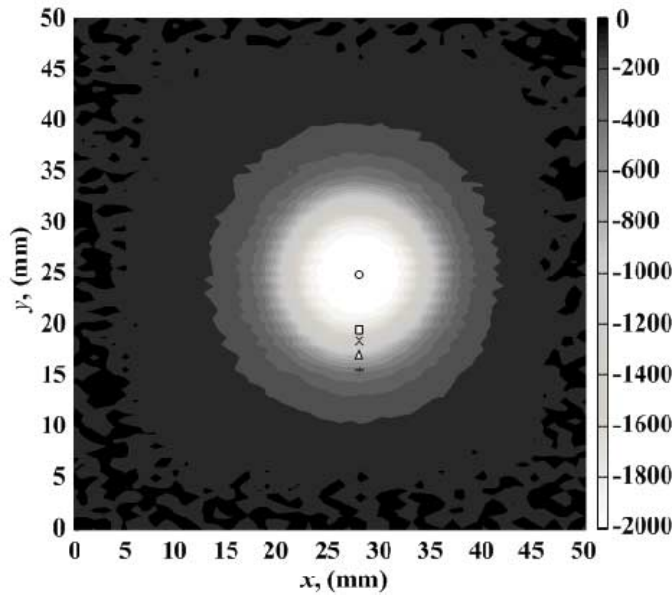


Figure 7. 2-D surface potential distribution at the beginning of the surface potential decay experiment ($t = 0$ h).

The surface charges accumulated on the material can drift in two paths, which are through the surface and the bulk to the grounded electrode. The Japan Aerospace Exploration Agency (JAXA) and Saitama University have done many works on the surface resistance and the volume resistance of spacecraft dielectric materials in vacuum, such as paints and cover glass. They found that the surface resistances of spacecraft dielectrics were several orders higher than their volume resistances [36]. Kyushu Institute

of Technology (KIT) also has done some works on the surface resistance and volume resistance of polyimide and showed that the surface resistance of polyimide was one order higher than the volume resistance [37]. The surface current density of the sample can be determined by the following equation.

$$j_s(x, y, t) = \frac{\nabla^2 \phi_s(x, y, t)}{R_s} \quad (14)$$

where, $j_s(x, y, t)$ is the surface current density at different location at time t in A/m^2 , R_s is the surface resistivity of the insulator.

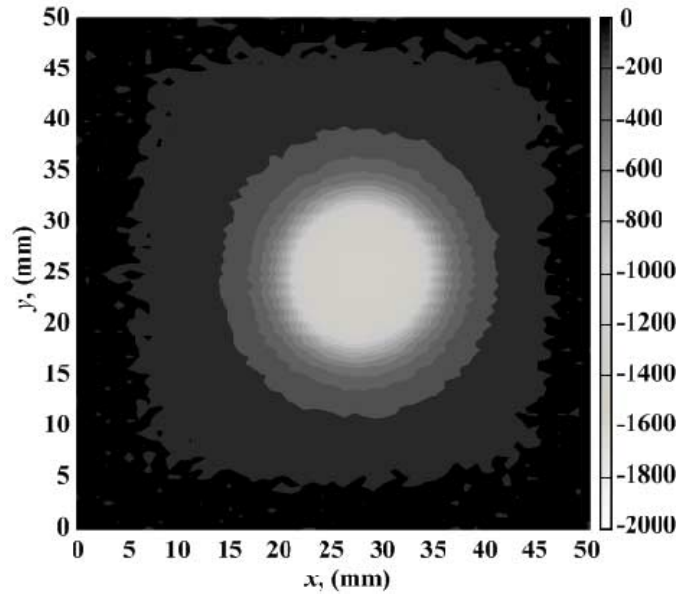


Figure 8. 2-D surface potential distribution at the end of the surface potential decay experiment ($t = 94.8$ h).

Since the surface current density was much smaller than the decay rate of surface charges through polyimide,

$$j_s(x, y, t) \ll \frac{dq_s(x, y, t)}{dt} \quad (15)$$

therefore, we neglected the surface charge transport in the present paper.

Analyzing the surface potential at a fixed point against time, we got the surface potential decay curves at different locations shown in Figure 9. At the beginning, the surface potential decayed very fast. The reason is the injected electrons from the deposited electron layer drift to the unoccupied traps. As the charges are filling the traps, their mobility is reduced and thus the potential decay slows down, as shown in the long time region in Figure 9.

In order to calculate the surface potential decay rate at different initial charging potential, we fitted the surface potential of the irradiated insulator by a power law [38]. After fitting the surface potential decay curves by the power function ($a \cdot t^b + c$, a , b , and c are constants), we obtained Figure 10, where the fitting errors, R^2 , were all less than 0.5%.

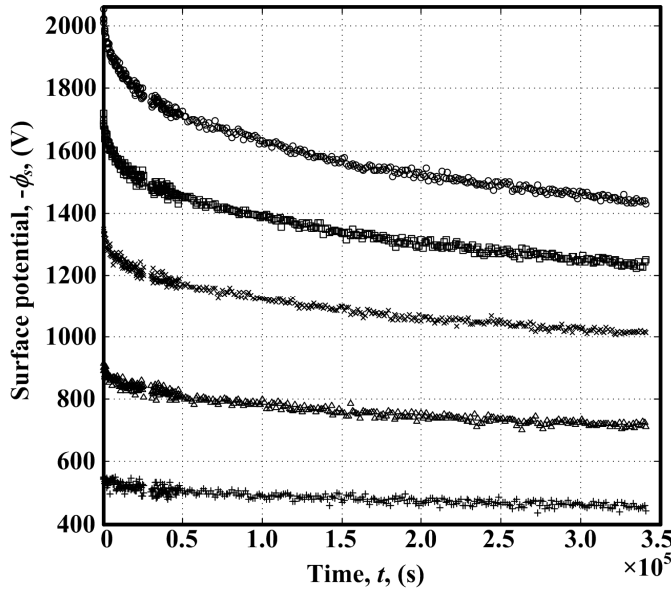


Figure 9. Experimental curves of surface potential decay at the various locations on the insulator. The coordinates of the five locations were indicated in Figure 7.

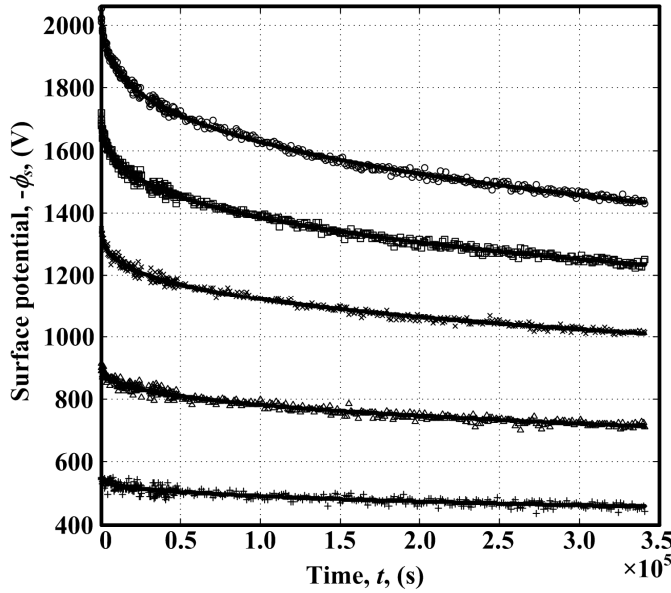


Figure 10. Fitted curves of surface potential decay at the various locations on the insulator. The experimental data are shown by marks and the fitting results are shown by solid curves. The fitting errors, R^2 , were less than 0.5%.

4.1 OHMIC RESISTIVITY

Since the relative permittivity of polyimide changes a little with temperature and electric field, we assumed it as a constant in this work [39]. The relative permittivity of polyimide was about 3.4 in the frequency range of 10^{-1} – 10^3 Hz at room temperature measured by Novocontrol broadband dielectric spectroscopy. Then, we obtained the surface potential decay rate curves, $\epsilon_0 \epsilon_r d\phi_s/dt$, shown in Figure 11. It shows that the higher the initial charging potential was, the larger the surface potential decay rate would be. It was assumed that the surface potentials decay rate at the steady state ISPD, such as $t=3 \times 10^5$ s, equaled to the conduction current density across the insulator.

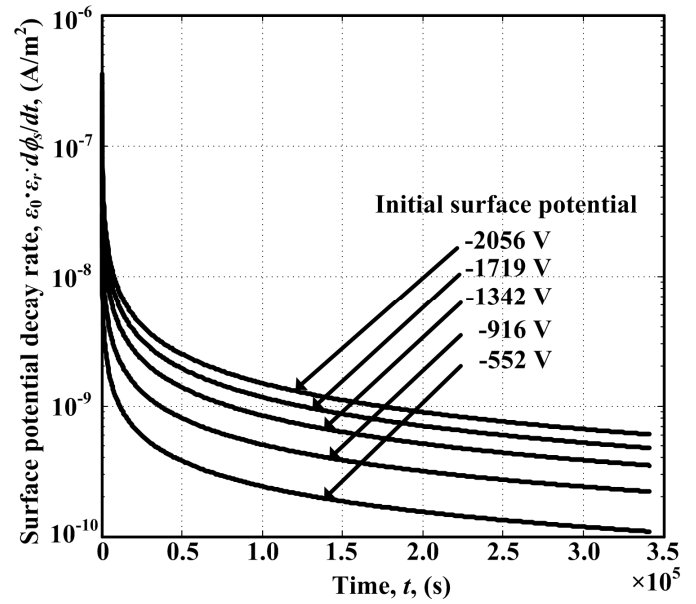


Figure 11. Surface potential decay rate characteristic at the various initial surface potentials.

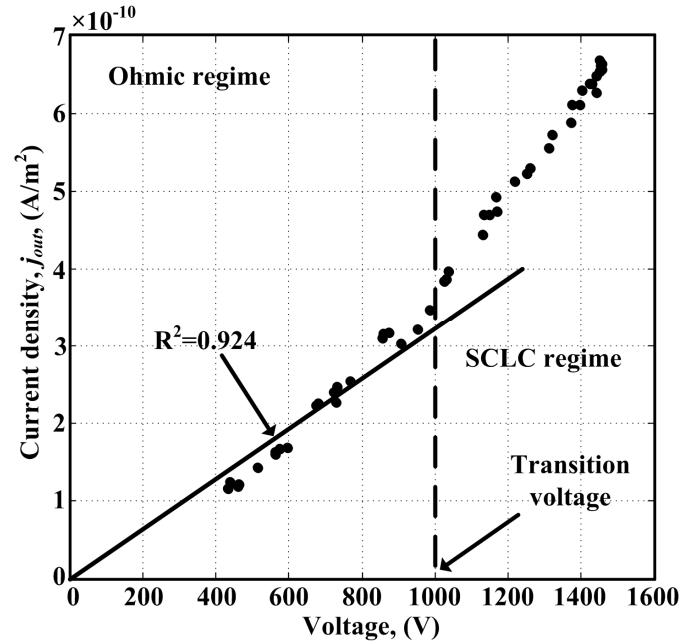


Figure 12. Voltage-Current density characteristic of virgin polyimide measured by the isothermal surface potential decay method at room temperature. The R square of a linear fit curve at Ohmic regime is 0.924. The voltages in the horizontal axis were that at the various locations of the sample at $t=3 \times 10^5$ s.

We plotted the steady state current density-voltage characteristics of polyimide, shown in Figure 12. It is clear that the steady state current density can be divided into two regimes, which are Ohmic current at low voltage and Space Charge Limited Current (SCLC) at high voltage depending on whether the applied voltage exceeds threshold or not [26, 27].

$$j_{out}^{steadystate}(V) = \begin{cases} j_{Ohmic}(V), & V < V_T \\ j_{SCLC}(V), & V > V_T \end{cases} \quad (16)$$

where, j_{Ohmic} is the Ohmic current density in A/m²; V is the applied voltage on the material in V; V_T is the transition voltage between the Ohmic regime and the SCLC regime in V.

The slope of a linear fit curve between the current density and the voltage at the Ohmic regime in Figure 12 was 3.11×10^{-13} . Therefore, we obtained the Ohmic resistivity of virgin polyimide from the Ohmic regime by equation (17),

$$\rho_{Ohmic} = \frac{V}{j_{Ohmic} \cdot L} \quad (17)$$

which was about $1.2 \times 10^{17} \Omega \cdot m$. The result is in the same order as that measured by charge storage decay resistivity test method [8, 40]. Moreover, this region indicates that the current density is too small to be measured by the ASTM method [9, 10].

4.2 CHARGE CARRIER MOBILITY

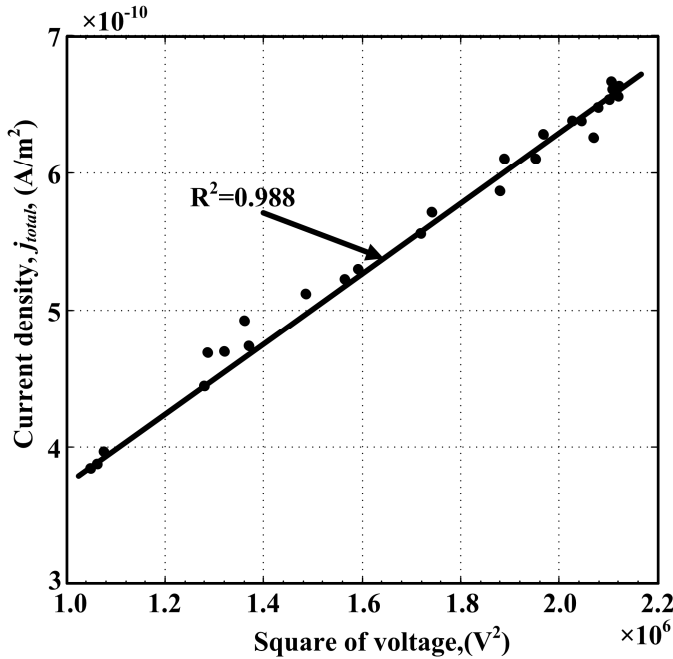


Figure 13. Space charge limited current characteristic of virgin polyimide measured by the isothermal surface potential decay method at room temperature. The R square of a linear fit curve of the SCLC regime is 0.988.

Redrawing the current density with the square of voltage applied on polyimide at the SCLC regime, we obtained Figure 13. The current density increased linearly with the square of voltage. The slope of a linear fit curve between the current density and the square of voltage at the SCLC regime was 3.25×10^{-16} . Consider the space charge limited current in dielectrics [41-43];

$$j_{SCLC} = \frac{9}{8} \cdot \epsilon_0 \cdot \epsilon_r \cdot \theta \cdot \mu \cdot \frac{V^2}{L^3} = \frac{9}{8} \cdot \epsilon_0 \cdot \epsilon_r \cdot \mu_{eff} \cdot \frac{V^2}{L^3} \quad (18)$$

where, μ_{eff} is the effective charge carrier mobility in m²/V·s.

We calculated the effective charge carrier mobility of polyimide by equation (18). The effective charge carrier mobility was $1.9 \times 10^{-19} \text{ m}^2/\text{V} \cdot \text{s}$. The charge carrier mobility of polyimide measured by G. M. Sessler *et al.* was 4×10^{-16}

m²/V·s at 323 K and $10^{-13} \text{ m}^2/\text{V} \cdot \text{s}$ at 473 K [13]. Since the carrier mobility decreases with the decrease of temperature, the results are in reasonable agreement with each other.

The transition voltage between the Ohmic regime and the SCLC regime could be calculated by the following equation.

$$j_{Ohmic}(V) = j_{SCLC}(V) \quad (19)$$

Then, we obtained the transition voltage, $V_T = -950 \text{ V}$.

4.3 TRAP DENSITY

When the initial potential was -2056 V, the surface potential decayed to -1456 V for the decay time of $3 \times 10^5 \text{ s}$, as demonstrated in Figure 9. When the surface potential at the steady-state ISPD was larger than the transition voltage, the current density versus voltage characteristic was in SCLC regime. For Child's law, M. A. Lampert found that the electric field at the grounded electrode (anode) satisfied the equality, $EF(L) = 3\phi_s/2L$ [26]. From the equality, we obtained $EF(L) = -8.09 \times 10^7 \text{ V/m}$ at the location with the initial surface potential of -2056 V. Accordingly, by substituting the electric field at the grounded electrode, the effective charge carrier mobility ($1.9 \times 10^{-19} \text{ m}^2/\text{V} \cdot \text{s}$), and the space charge limited current density ($6.63 \times 10^{-10} \text{ A/m}^2$) into equation (13), we obtained the trap density of polyimide, which was about $1.3 \times 10^{21} \text{ m}^{-3}$. The result is consistent with that measured in [14, 18].

5 CONCLUSIONS

Isothermal Surface Potential Decay (ISPD) experiments were carried out to investigate the transport properties of polyimide used on spacecraft. Ignoring the charges leakage due to the surface conductance, we calculated the current density at various initial surface potentials at the steady state ISPD. We found that the voltage-current characteristic of polyimide could be separated into two different regimes at about -950 V with the sample thickness of 27 μm , which were the Ohmic regime at low electric field and the SCLC regime at high electric field. We obtained the Ohmic resistivity and the effective charge carrier mobility from the two regimes, which were $1.2 \times 10^{17} \Omega \cdot m$ and $1.9 \times 10^{-19} \text{ m}^2/\text{V} \cdot \text{s}$, respectively. Considering the current density versus voltage characteristic was SCLC at steady state ISPD with high initial surface potentials, we developed the ISPD model to get more details of the charge transport properties. The trap density of polyimide was calculated by using this model, which was $1.3 \times 10^{21} \text{ m}^{-3}$.

Temperature dependent Ohmic resistivity and charge carrier mobility will be studied by ISPD experiments to get better comprehension of the charge transport of high insulation materials. These charge transport parameters will be used to update the parameters in Multi-Utility Spacecraft Charging Analysis Tool (MUSCAT) to get the more accurately spacecraft charging and discharging simulating results in the complex space environment [44, 45]. The results, then, will provide us more details to find methods to improve the ESD mitigation technology. In the near future, we will also use these charge transport parameters to develop a secondary electron yield measurement method.

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